Phase5 Test Plan

Moderated Learning Management System (LMS)

Phase 5 — Verification and Validation

5.1 Test strategy and V-Model traceability

V-Model test level	Design/development artefact verified	Main techniques	Entry criteria	Exit criteria
Unit	Business rules, class methods, validation rules and LatePenaltyStrategy	Equivalence partitioning, boundary-value analysis, positive/negative tests	Requirement and method contract reviewed; build succeeds	All planned unit cases pass; no open critical/high defects; results recorded
Integration	Repository/data-access layer, SQLite schema, service-to-repository interactions, OpenSSL integration	CRUD workflow tests, transaction/constraint/error-path tests, role-permission tests	Unit tests pass; test database is isolated; schema initialises	Data persists and constraints hold; prepared statements work; all integration cases pass
System	Complete use-case flows across Student, Lecturer, HoD and Administrator	End-to-end scenarios, exception-flow tests, role misuse tests	Integrated build and test data available; use cases baselined	All critical workflows pass; defects triaged; no critical/high open defects
Acceptance	SRS Must requirements and user stories/use-case specifications	User-acceptance scripts, checklist and stakeholder sign-off	System tests pass; release candidate and user manuals available	School representative confirms required workflows and records acceptance or rejection

5.2 Equivalence partitions and boundary-value analysis

The current late rule is 10 percentage points of the assignment maximum mark per late day, capped at three late days; submissions more than three days late are rejected by the service. A valid extension changes the effective deadline. These partitions are used:

Input	Equivalence partitions	Boundary values / expected result
Submission time	Before deadline; exactly at deadline; after deadline	deadline − 1 minute = no penalty; exact deadline = no penalty; deadline + 1 minute = 1 late day under the current day-ceiling rule
Days late	0; 1; 2–3; more than 3	0 = raw mark; 1 = raw − 10% max; 3 = raw − 30% max; 4 = rejected by submit service / strategy returns zero
Extension	No extension; individual extension; class extension; invalid/empty reason	Effective deadline is the latest applicable extension; empty reason rejected
Mark	Negative; 0; within range; maximum; above maximum	0 and maximum accepted; negative and above maximum rejected
Password	Fewer than 8 characters; 8+ characters; correct; incorrect	7 rejected; 8 accepted; correct verifies; incorrect fails
Role	Correct role; wrong role; inactive account	Only active authorised role can perform the action

Clarification: the strategy's daysLate calculation rounds any positive lateness up to a whole day. The school should approve this interpretation; if it intends calendar-day rather than rolling-24-hour penalties, the rule and tests must be changed consistently.

5.3 Automated tests

- tests/test_lms.cpp: dependency-free CTest smoke/integration executable, retained so a local build does not need a test-framework download.
- tests/test_lms_gtest.cpp: GoogleTest suite for CI, enabled with -DLMS_ENABLE_GTEST=ON. It covers password hashing, penalty boundaries, and CR1 class-wide extension behaviour, including two enrolled students and denied/invalid requests.
- CI configures, builds and runs both suites. GoogleTest is fetched by CMake during CI configuration.

Run locally without downloading GoogleTest:

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Debug
cmake --build build
ctest --test-dir build --output-on-failure
```

Run the GoogleTest suite where internet access is available:

```
cmake -S . -B build-gtest -G Ninja -DLMS_ENABLE_GTEST=ON
cmake --build build-gtest
ctest --test-dir build-gtest --output-on-failure
```

5.4 System and acceptance scenarios

ID	Use case / scenario	Expected result	Evidence/status
SYS-01	Administrator creates accounts, department, course and enrolment	Records persist in SQLite; role types are retained	Automated integration test PASS
SYS-02	User logs in with correct and incorrect password	Correct password authenticates; wrong password does not	Automated test PASS
SYS-03	Lecturer grants class extension to an assignment	All enrolled students receive new effective deadline	CR1 integration test PASS for two students
SYS-04	Administrator attempts to grant an extension	Access denied	CR1 integration test PASS
SYS-05	Lecturer grants extension without reason	Validation exception; no extension inserted	CR1 integration test PASS
SYS-06	Student attempts submission beyond allowed period	Submission rejected with an explanatory error	Manual acceptance script; execute against a prepared course before school sign-off
SYS-07	Lecturer marks; HoD moderates; HoD releases	Mark transitions through pending/approved/released; audit row recorded	Manual acceptance script; execute with test accounts before school sign-off
SYS-08	Student views assignment list after class extension	Assignment list displays effective deadline and EXTENDED indicator	UI path implemented; verify interactively before acceptance sign-off

The automated tests were run in the development container on 2 November 2026: CMake build succeeded and CTest reported 1/1 test passed. GoogleTest CI tests are defined but require the CI runner to fetch GoogleTest; they are not claimed as locally executed. The manual school-acceptance scenarios SYS-06 to SYS-08 remain pending witnessed execution.

5.5 Defect log

Defect	Severity	Cause	Fix	Retest
D-01: no whole-class extension operation	Medium	Extension table/API represented only per-student grants	Added ClassExtensions, repository/service method and lecturer menu option	PASS: two students receive class deadline
D-02: student list did not identify extended deadlines	Low	UI printed date without distinguishing an extension	Student assignment list now appends (EXTENDED) when effective deadline differs from base deadline	Implemented; manual visual check pending
D-03: extension reason could be empty	Medium	Repository only checked reason on individual extensions	Both individual and class extension methods reject empty reason	PASS: automated negative test
D-04: non-lecturer could call class extension service	High	New operation needed a service-layer permission gate	Requires active lecturer and assignment ownership	PASS: administrator-denial test

5.6 Coverage report

A GCC coverage-instrumented build was run using --coverage -O0; CTest passed 1/1. gcov reported 43.57% line execution (183/420 executable lines) for the instrumented lms.cpp compilation unit in this run. This is the actual measured figure, not the 70% target. The target is therefore not yet achieved. Additional tests are needed for moderation/release, mark audit, file submission/resubmission, repository error paths, and every role-permission branch. Coverage is compiler/build dependent and should be regenerated by CI on the target commit.

To reproduce:

```
cmake -S . -B build-coverage -G Ninja -DCMAKE_BUILD_TYPE=Debug \
  -DCMAKE_CXX_FLAGS="--coverage -O0" -DCMAKE_C_FLAGS="--coverage -O0" \
  -DCMAKE_EXE_LINKER_FLAGS="--coverage"
cmake --build build-coverage
ctest --test-dir build-coverage --output-on-failure
cd build-coverage
gcov -b -c CMakeFiles/lms_core.dir/src/lms.cpp.o
```

References

The following sources support the standards, tools and practices discussed in this document. URLs were checked on 2 November 2026.

1. International Organization for Standardization. (2022). ISO/IEC/IEEE 29119-1:2022 — Software and systems engineering — Software testing — Part 1: General concepts. https://www.iso.org/standard/81291.html
2. International Organization for Standardization. (2021). ISO/IEC/IEEE 29119-2:2021 — Software testing — Part 2: Test processes. https://www.iso.org/standard/79428.html
3. International Organization for Standardization. (2021). ISO/IEC/IEEE 29119-3:2021 — Software testing — Part 3: Test documentation. https://www.iso.org/standard/79429.html
4. Hamburg, M., & Roman, A. (2025). Boundary Value Analysis According to the ISTQB Foundation Level Syllabus. ISTQB / German Testing Board / Polish Testing Board. https://istqb.org/wp-content/uploads/2025/10/Boundary-Value-Analysis-white-paper.pdf
5. Google. (n.d.). GoogleTest Primer. GoogleTest User's Guide. https://google.github.io/googletest/primer.html